MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE DOD-001 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Reflex, integral cavity, mechanically tunable

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Ik	Ehk	F	TA
Unit:	V	Vdc	Vdc	mA	Vdc	MHz	°C
Maximum:	6.8	330	-25	36	+100	---	100
Minimum:	5.7	---	-380	---	-100	---	---

(see note 1)

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Cathode:	Unipotential

TEST CONDITIONS:

Parameter:	Ef	Ers	Er	F
Unit:	V	Vdc	Vdc	MHz
Test 1:	6.3	300	2-3/4 (mode)	7,125 to 7,770
Test 2:	6.3	300	3-3/4 (mode)	7,050 to 7,845

GENERAL:

Qualification - Required

1/ Formerly QKK-752

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 (see note 6)					
4214	Cathode emission	Test condition 1; Ef = 5.8 V; tk = 120; F = 7,125 MHz	ΔIk/Ik	---	15	%
1256	Electrode current (cathode)	Test condition 1; F = 7,125 MHz	Ik	---	32	mAdc
4250	Power output (1)	Test condition 1; Er/max Po (see notes 2 and 3)	Po	90	---	mW
4250	Power output (2)	Test condition 2; Er/max Po (see notes 2 and 3)	Po	50	---	mW
4213	Reflector voltage (1)	Test condition 1; F = 7,125 MHz (see notes 2 and 3)	Er	-190	-290	Vdc
4213	Reflector voltage (3)	Test condition 2; F = 7,050 MHz (see notes 2 and 3)	Er	-60	-150	Vdc
4229	Total reflector current	t = 120	Ir	---	7	μAdc
4229	Reflector-leakage current		Ir	---	5	μAdc
4229	Reflector-gas current		Ir	---	2	μAdc
---	Reflector modulation sensitivity (1)	Test condition 1; Er/max Po; 60 Hz modulation voltage; 10 V (p-p); F = 7,125, 7,450, and 7,770 MHz (see note 2)	ΔF/ΔEr	0.2	0.95	MHz/V
	Quality conformance inspection, part 2					
1031	High-frequency vibration	t = 120; 10 G; F = 50; no voltages	---	---	---	---
1301	Heater current	Ef = 6.3 V	If	400	480	mAdc
1336	Heater-cathode leakage	Ehk = ±100 Vdc	Ihk	---	100	μAdc
1211	Insulation of electrodes	Tube cold Ek-rs = 300 Vdc Eh-rs = 300 Vdc	 Ik-rs Ih-rs	 --- ---	 150 150	 μAdc μAdc
4213	Reflector voltage (2)	Test condition 1; F = 7,770 MHz (see notes 2 and 3)	Er	-260	-360	Vdc
4213	Reflector voltage (4)	Test condition 2; F = 7,845 MHz (see notes 2 and 3)	Er	-90	-200	Vdc
4280	Electronic tuning range	Test condition 1; Er/50% max Po; F = 7,125, 7,450, and 7,770 MHz (see note 2)	ΔF	15	---	MHz

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 2 - Continued					
---	Reflector modulation sensitivity (2)	Test condition 2; Er/max Po; 60 Hz modulation voltage; 10 V (p-p); F = 7,050, 7,440, and 7,845 MHz (see note 2)	ΔF/ΔEr	0.50	3.0	MHz/V
---	Long-term frequency stability	Test condition 1; F = 7,770 MHz (see note 4)	ΔF	---	±3	MHz
4027	Temperature coefficient	Test condition 1; F = 7,770 MHz (see note 5)	ΔF/ΔC	---	±0.2	MHz/°C
	Quality conformance inspection, part 3					
---	Life-test provisions	Group D; test condition 1	t	2,000	---	hrs
---	Life-test end point	Power output (1)	Po	70	---	mW

NOTES:

1. The temperature of the output lead shall not exceed 90°C.
2. The tube shall be mounted in a suitable socket and shall be fixed firmly by means of clamps which make contact with the tube only at the flange adjacent to the base. Measurements of the tube in an oscillating state shall be made with the output lead coupled into a launching section as indicated on figure 1, or equal.
3. Reflector voltage and shorting plunger of the starting section shall be mutually adjusted for maximum power output.
4. The tube is set at a frequency of 7,770 MHz and allowed to warm-up for 4 hours in a heat chamber. The temperature of the circulating air within the chamber should be 65° ±5°C initially with a maximum variation of ±2°C throughout the test. During the next 8 hours of running time, the frequency should be read a minimum of five times, no readings to be taken within 1 hour of each other. The deviation of the first five or more readings from the average should not be greater than ±3 MHz and the readings should be averaged to obtain an initial tube frequency.

   The tube is then allowed to run another 88 hours without taking any readings.

   During the next 8 hours, which gives a total of 104, the frequency is again read a minimum of five times with no two readings occurring within 1 hour of each other. These five readings should be averaged and this average should be within ±3 MHz from the initial tube frequency determined from the first five readings.

   All readings should be made at the same resonator, reflector and heater voltages.
5. The tube shall be operated in an insulated chamber which provides a high degree of isolation to random changes in room ambient temperature. An output waveguide shall be provided for measurement of operating frequency. The temperature in the chamber shall be maintained using a heat source controlled by a thermostat capable of regulating the temperature within ±2°C. A fan shall be used to insure uniform temperature within the chamber. The chamber temperature shall be monitored using a thermometer having an accuracy within ±1°C. The reflector voltage shall be adjusted for maximum power output at the test frequency and maintained within 0.10 volts thereafter. The chamber ambient temperature shall be adjusted through 3 cycles from low temperature, T = 40°C, to high temperature, T = 100°C, and the tube frequency shall be measured at low and high temperatures for each cycle. After the

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chamber has reached the operating temperature, sufficient time shall be allowed to achieve stabilization of the tube frequency. The initial chamber temperature shall be 100°C. The temperature coefficient shall be determined as the difference between the average of the three frequency readings at 100°C and the average of the three frequency readings at 40°C divided by 60.

6. The combined AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 with an inspection level of II.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL
Navy -
Air Force - 11, 71, 85
DSA - ES

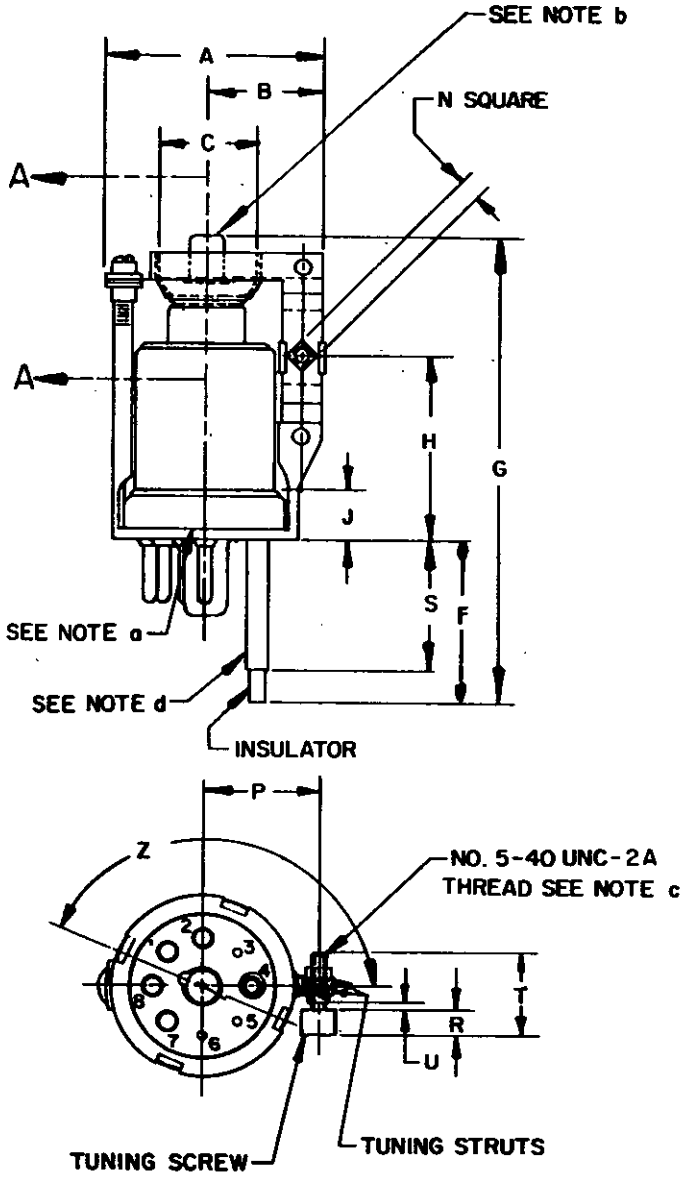
User activities:
Army - MU, ME, WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2397)

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PIN NO.	ELEMENT
1	SHELL
2	HEATER
4	OUTPUT LEAD
7	HEATER
8	CATHODE

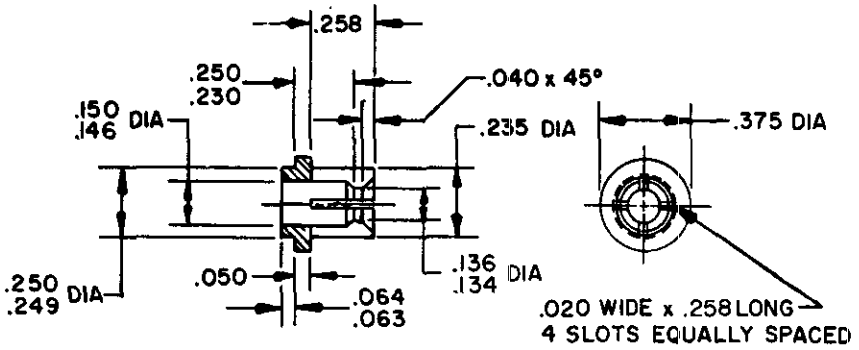
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
H	1.313 (33.35)	1.437 (36.50)
P	.656 (16.66)	.719 (18.26)
Z	154°30′	160°30′
Quality conformance inspection, part 2		
A		1.609 (40.87)
G		3.562 (90.47)
N	.182 (4.62)	.192 (4.88)
R	.203 (5.16)	.234 (5.94)
U		.135 (3.43)
Quality conformance inspection, part 3 (periodic check)		
B		.859 (21.82)
C	.767 (19.48)	.797 (20.24)
F	1.203 (30.55)	1.235 (31.35)
J	.330 (8.38)	.400 (10.16)
S	.875 (22.23)	.985 (25.02)
T	.804 (20.42)	.835 (21.21)

NOTES:

a. Base: B6-92. (except dimension 'S')
b. Cap: C1-4.
c. The tuning screw shall be lubricated with oildag or equivalent non-corrosive lubricant - it shall be capable of being operated smoothly through its entire range without perceptable binding.
d. Nickel (30 MSI silver plate permissible).
e. Applies to centerline of output lead and centerline of tuner through struts.

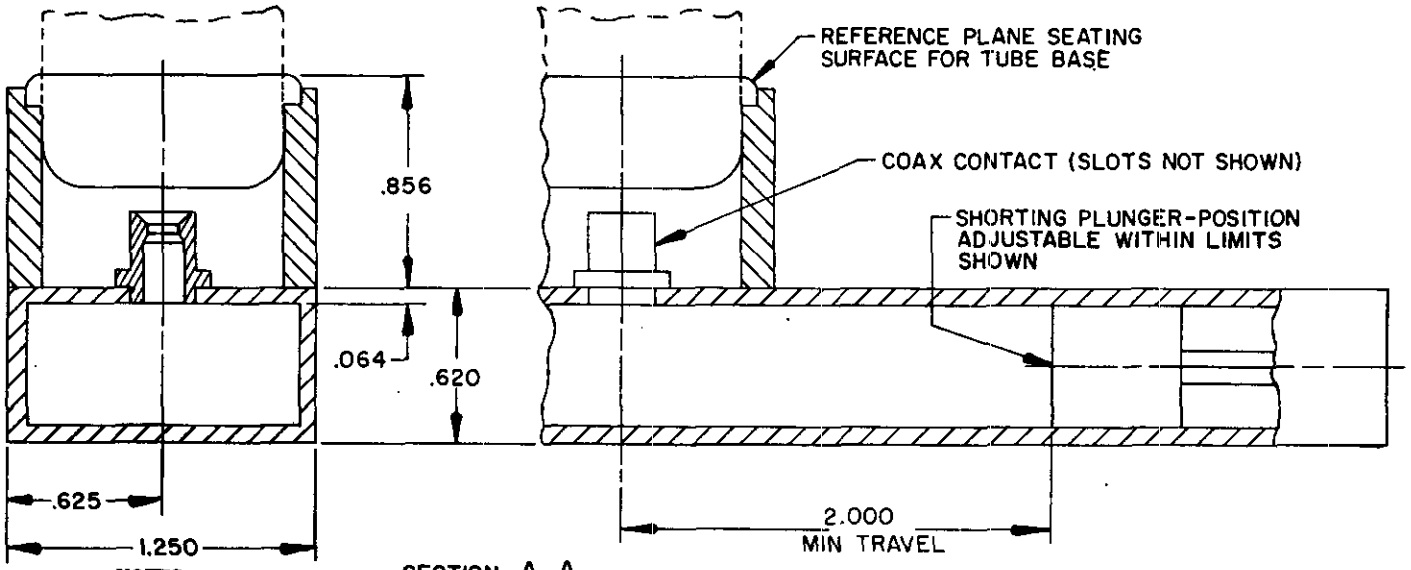
FIGURE 1. Outline drawing of electron tube type ROD-001

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INCHES	MM	INCHES	MM
.020	.51	.230	5.84
.040	1.02	.235	5.97
.050	1.27	.249	6.32
.063	1.60	.250	6.35
.064	1.63	.258	6.55
.120	3.05	.375	9.53
.134	3.40	.620	15.75
.136	3.45	.625	15.88
.149	3.78	.856	21.74
.150	3.81	1.250	31.75
.186	4.72	1.750	44.45

A. All dimensions are given as finished. Allow .0005 on each surface for silver plating.
B. Beryleo No. 25 or No. 40 heat treat to harden 600°F - 2 hr(or equivalent material and process).



SECTION A-A

NOTES:
a. Dimensions are in inches.
b. Unless otherwise specified, tolerances are ±.005 (.13 mm).

FIGURE 1. Outline drawing of electron tube type DOD-001 - Continued.

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